

Retirement Simulation Engine: Architecture, Mathematical Mechanics & Critical Evaluation

A Complete Guide for Non-Technical Stakeholders on How Probabilities, Taxes, Withdrawals, Asset Growth, and Historical Stress Tests Are Calculated

Original User Request / Prompt:

"Describe to me, in terms that someone who is not familiar with coding or the mathematics of statistics, (a) how the probability of success is determined, (b) how the historical stress tests are calculated, (c) how each of the charts under the charts tab is calculated (each when showing nominal values and when showing real values), (d) how taxes and penalties are calculated and used in the app (including how taxes on social security are calculated), (e) how portfolio withdrawals are calculated and used in the app, and (f) how asset growth is calculated and used in the app. Although your descriptions should be in terms that someone who is not familiar with the mathematics of statistics can understand, the descriptions should be accurate, complete, and not oversimplified. Please describe any criticisms, errors, omissions, and weaknesses of any of the foregoing. If possible, please save your response (and this prompt) in a pdf document saved to my github/retire folder. If you cannot save it as a pdf document, please save it in another format that will be readable by a person who is not a coder or a person familiar with the mathematics of statistics and that will print in a readable format."

Executive Summary & Core Philosophy

This application is a specialized financial planning simulation engine built to project household wealth across retirement. Unlike basic calculators that simply assume your money earns a steady 6% every single year, this application evaluates both **Deterministic Projections** (a single baseline pathway using average expected returns) and **Monte Carlo Simulations** (thousands of alternate economic scenarios reflecting real-world market randomness and volatility).

The model is architected around an annual cash-flow simulation cycle: for every year of life from your current age to your planned age of death, the application computes all incoming revenues (Social Security, pensions, annuities), deposits new savings, applies investment returns across distinct asset accounts, checks Required Minimum Distributions (RMDs), deducts living expenses, calculates federal and state taxes using statutory brackets, and follows a strict withdrawal priority if your income falls short of your spending needs.

Section (a): How the Probability of Success Is Determined

In plain language, the **Probability of Success** represents the percentage of simulated alternate lifetimes in which your investment portfolio does not run out of money before you (and your surviving spouse) pass away.

1. The 10,000 Lifetime Experiment:

By default, the simulation creates **10,000 distinct lifetimes** (simulation runs). For every lifetime, the computer generates a fresh sequence of random annual market returns for stocks, bonds, and cash spanning your entire retirement horizon. In some lifetimes, the stock market enjoys a boom during your early retirement years; in others, severe recessions strike right as you stop working (a phenomenon known as 'Sequence of Returns Risk').

2. Tracking the Portfolio Balance:

Within each individual simulated path, your accounts are tracked year-by-year. In years when your living expenses and taxes exceed your income, money is pulled from your portfolio. If your portfolio accounts are completely drained to zero and cannot cover your remaining expenses, the unpaid shortfall accumulates as a negative balance in your taxable account (representing debt or unmet spending).

3. The Terminal Test & Mathematical Formula:

At the final year of the simulation (your designated age of death, or the surviving spouse's age of death), the simulator checks the final ending portfolio balance (excluding any life insurance death benefit paid to heirs). Each simulation run receives a binary grade:

- **SUCCESS (1.0):** If Ending Portfolio Wealth $\geq$ \$0.
- **FAILURE (0.0):** If Ending Portfolio Wealth $<$ \$0.

The overall Probability of Success is simply the total number of successful runs divided by the total number of simulations, multiplied by 100:

$$\text{Probability of Success (\%)} = \left(\frac{\text{Number of Runs Ending with Wealth} \geq \$0}{\text{Total Simulation Runs}} \right) \times 100\%$$

4. Goal-Seeking Mode (Maximum Spending):

When you select 'Maximum Spending' mode, the application reverses the question. Instead of asking 'What is my chance of success with \$50,000 spending?', it asks: 'What is the maximum annual spending I can afford while maintaining an 80% (or other target) probability of success?' The application uses an efficient **binary search algorithm**: it tests a spending level across all 10,000 runs, checks the resulting success rate, and iteratively cuts the search window in half over 25 rounds until it identifies the exact annual spending level that achieves your target success rate within pennies.

CRITIQUE & WEAKNESS: PROBABILITY OF SUCCESS

Critical Flaw — The 'Debt Erasure' Blind Spot:

In this codebase (*runs.py*, line 1622), success is judged **only at the final terminal year**. If a retiree runs out of money in year 10, their taxable account balance goes negative. The software charges 0% interest on this deficit. If in year 20 a large inheritance, sale of a home, or delayed Social Security benefit arrives that wipes out the debt before the final year of life, the application marks that lifetime as a **100% SUCCESS!** In the real world, a retiree cannot borrow hundreds of thousands of dollars interest-free to survive a decade of insolvency. Institutional tools define failure as hitting zero at *any* point in the retirement timeline.

Section (b): How the Historical Stress Tests Are Calculated

While Monte Carlo simulations evaluate generic statistical randomness, the **Historical Stress Test** confronts your plan with the brutal realities of actual American economic history. It tests whether your portfolio could survive the greatest financial catastrophes of the past century.

1. The Crisis Scenarios:

The application includes monthly and annual historical data from Robert Shiller (Yale), Ibbotson SBBI, and FRED spanning 1926 through 2024. Six specific crisis periods are pre-configured:

- **The Dot-Com Crash & Lost Decade (2000–2012, 13 years):** Three straight years of negative stock returns (-9.1%, -11.9%, -22.1%) followed by the 2008 crash.
- **The 1970s Stagflation & Inflation Shock (1973–1982, 10 years):** Stock declines (-14.7%, -26.5%) combined with double-digit inflation reaching 13.3%.
- **The Great Recession (2007–2009, 3 years):** A 37% stock crash in 2008 offset by a +20% flight-to-safety surge in Treasury bonds.
- **The Great Depression (1929–1941, 13 years):** Four consecutive years of catastrophic market collapse (-8.4%, -24.9%, -43.3%, -8.2%) with severe deflation.
- **The 2022 Inflation & Rate Spike Shock (2022–2024, 3 years):** Rare simultaneous drop in both stocks (-18.1%) and bonds (-17.8%) as inflation peaked.
- **The 1966 Sideways Market & Inflation (1966–1981, 16 years):** The historical worst-case retirement cohort in Bill Bengen's original 4% rule study.

2. The Hybrid Simulation Engine:

The stress test runs a **hybrid Monte Carlo simulation**. You choose when the crisis strikes: either 'At Retirement Start' or 'Immediately at Plan Inception'. During the designated crisis window (e.g., the 13 years of the Dot-Com crash):

1. Random returns are completely deactivated.
2. Every single one of the 10,000 simulation runs is forced to experience the **exact historical calendar returns** for stocks, bonds, and cash, as well as the **exact historical CPI inflation rates** of that era.
3. Account returns are blended based on your asset allocation (either fixed presets like 60/40 or automatically inferred from your expected returns).
4. For all years outside the crisis window (before the crisis begins, and after it concludes until death), the simulation returns to standard random Monte Carlo modeling.

The software then calculates the resulting **Stress Success Rate**, compares it to your baseline success rate, and reports the **Delta (Impact)** across all ending wealth percentiles.

CRITIQUE & WEAKNESS: HISTORICAL STRESS TESTS

Criticisms & Weaknesses of the Stress Test:

- **Wrap-Around Artifacts:** In *historical_data.py* (line 181), if a crisis simulation extends beyond 2024, the calendar counter loops directly back to 1926! Testing a crisis starting in 2022 would transition from 2024 market returns straight into the Roaring Twenties.
- **Static Identical Shock:** Because the exact historical returns are deterministic, every one of the 10,000 runs suffers the exact same crisis sequence. Only the post-crisis years experience variation. It does not test synthetic crises (e.g. what if 2008 had lasted 5 years instead of 1 year?).

Section (c): How Each Chart Under the Charts Tab Is Calculated

Under the **Charts** tab, five interactive visual tools illustrate your financial future. Users can toggle between **Nominal Dollars** (future dollars as they will appear on bank statements) and **Real Dollars** (inflation-adjusted dollars showing today's purchasing power).

The Nominal vs. Real Scaling Formula:

To convert any nominal dollar amount at year t into today's purchasing power, the software divides by the cumulative inflation factor: **Real Value = Nominal Value $\div$ (1 + Inflation Rate) ^{t}** . For ending asset balances, because they are measured at the end of the year, the exponent used is $(t + 1)$.

Chart Name	What It Shows & Calculation Source	Nominal vs. Real Mechanics
1. Spaghetti Chart	Plots up to 500 individual simulated Monte Carlo paths simultaneously across time. Each line is one specific simulated lifetime showing portfolio wealth.	• Nominal: Raw simulated portfolio wealth. • Real: Each year's wealth divided by $(1+\text{inf})^t$.
2. Monte Carlo Wealth Trajectory	Summarizes all simulation runs into 3 statistical benchmarks across time: • 90th Percentile (Optimistic): Top 10% market run. • 50th Percentile (Median): Middle run. • 10th Percentile (Pessimistic): Unlucky market run.	• Nominal: The 90th, 50th, and 10th percentiles of raw wealth at each year. • Real: Percentiles discounted by $(1+\text{inf})^t$.
3. Stacked Asset Class Breakdown	Stacked area chart from the <i>Deterministic Projection</i> showing portfolio composition over time across 4 account categories: • Pre-tax (IRA/401k) [Red] • Roth [Yellow] • Taxable [Blue] • HSA [Teal]	• Nominal: Ending balances of each account at year close. • Real: Each account's ending balance discounted by $(1+\text{inf})^{t+1}$.
4. Annual Income vs. Spending	Dual-stacked bar chart from the <i>Deterministic Projection</i> comparing cash flow: • Inflows: Social Security, Pensions, Portfolio Withdrawals. • Outflows: Regular Spending, Milestone Expenses, Taxes & Penalties.	• Nominal: Actual cash received and paid in year t. • Real: Cash flows discounted by $(1+\text{inf})^t$ to reflect today's living standard.
5. Lifetime Tax Liability & 'Tax Bomb'	Bar chart from the <i>Deterministic Projection</i> showing annual taxes paid (Federal + State + Penalties + Other). Bars switch from red to gold during years when RMDs begin.	• Nominal: Total tax liability paid in year t. • Real: Tax bills discounted by $(1+\text{inf})^t$ to reflect real purchasing power sacrificed to taxes.

CRITIQUE & WEAKNESS: CHARTS TAB

Criticisms & Weaknesses of the Charts:

- **Charts 3, 4, and 5 Are Deterministic Only:** While Charts 1 and 2 show Monte Carlo volatility, Charts 3, 4, and 5 depict a single deterministic average run. This can be deceptive: a retiree might see Chart 5 and assume their tax liability is fixed, whereas in volatile markets, tax liabilities swing wildly.
- **Constant Inflation Assumption:** Real mode discounts by a fixed exponential rate. If actual inflation spikes (as in 1973 or 2022), purchasing power degrades much faster than the smooth curve displayed.

Section (d): How Taxes and Penalties Are Calculated and Used

Taxes are one of the largest expenses in retirement. The application incorporates a fully integrated tax engine that models scheduled post-2025 federal income tax brackets, state income taxes, early withdrawal penalties, and the complex federal formula for taxing Social Security.

1. Federal Income Tax Brackets (Scheduled Post-2025 Law):

The software implements the scheduled reversion of the Tax Cuts and Jobs Act (TCJA) beginning in tax year 2026. It uses the standard progressive rate structure: 10%, 12%, 22%, 24%, 32%, 35%, and 37%. Standard deductions start at \$16,100 for Single filers and \$32,200 for Married Filing Jointly. Every year into the future, the dollar thresholds for tax brackets and standard deductions are indexed upward by the inflation rate: $\text{Threshold}_t = \text{Base Threshold} \times (1 + \text{Inflation Rate})^t$.

2. Social Security Benefit Taxation (The IRS Provisional Income Formula):

The application strictly implements IRS Publication 915 rules to determine what portion (0%, 50%, or 85%) of your Social Security is subject to income tax:

- **Step 1: Calculate Provisional Income:** Provisional Income = Other AGI (pensions, RMDs, taxable withdrawals) + 50% of your annual Social Security benefits.
- **Step 2: Compare to Statutory Thresholds:**
 - *Married Filing Jointly:* Base Threshold = \$32,000; Upper Threshold = \$44,000.
 - *Single:* Base Threshold = \$25,000; Upper Threshold = \$34,000.

(Critical Note: By federal law, these statutory thresholds are NOT indexed to inflation!)
- **Step 3: Progressive Taxation Tiers:**
 - If Provisional Income is below the Base Threshold, **0%** of Social Security is taxable.
 - If between Base and Upper Threshold, up to **50%** of the excess is added to taxable income.
 - If above Upper Threshold, **85%** of the excess is added, up to a maximum cap of **85% of total benefits**.

3. State Income Tax & The Social Security Exemption:

State tax is computed as a flat percentage of state taxable income. The user can toggle whether their state exempts Social Security benefits. If exempt, Social Security is stripped out before applying the state tax rate; if non-exempt, the federally taxable portion is taxed at the state rate.

4. Early Withdrawal & Non-Medical Penalties:

- **Early Pre-Tax Penalty (10%):** If extra pre-tax distributions are required before age 59.5, the app levies an additional 10% federal penalty.
- **HSA Non-Medical Penalty (20%):** If an HSA account is designated for non-medical expenses, any distribution before age 65 triggers ordinary income tax PLUS a 20% penalty. At age 65 and older, the 20% penalty disappears, but distributions remain subject to ordinary income tax (functioning like a traditional IRA).

5. The Circular Tax Problem & The Gross-Up Solver:

In retirement planning, if you need \$10,000 of net cash to pay bills, you cannot simply withdraw \$10,000 from a 401(k). Withdrawing \$10,000 increases your taxable income, which increases your tax bracket, forces more of your Social Security to become taxable, and might trigger early penalties. If you only withdrew \$10,000, you would still be short on cash. The app solves this circular dependency using a **25-step binary search gross-up algorithm** (*runs.py, line 279*). It precisely calculates the exact gross pre-tax withdrawal necessary so that after all resulting federal taxes, state taxes, Social Security tax increases, and penalties are paid, the remaining net cash exactly satisfies your deficit.

CRITIQUE & WEAKNESS: TAXES & PENALTIES

Criticisms & Omissions in the Tax Engine:

- **Capital Gains Tax Omission:** Withdrawals from the taxable brokerage account are assumed to have **zero capital gains tax** (treated entirely as non-taxable return of principal). In reality, selling stocks or mutual funds triggers capital gains taxes and dividend drag.
- **Missing Medicare IRMAA Surcharges:** High taxable income or large RMDs push retirees into higher Medicare Part B & D premiums (IRMAA), creating hidden tax cliffs that this model omits.
- **The 'Widow's Penalty' Cliff:** In the exact year after the first spouse passes away, the surviving spouse's tax status drops immediately to Single. While realistic, the app omits the 2-year Qualifying Surviving Spouse relief.

Section (e): How Portfolio Withdrawals Are Calculated and Used

Every year, the simulator reconciles cash inflows and outflows to determine whether you have surplus savings or need to liquidate investments.

1. Determining Annual Surplus or Deficit:

- **Cash Inflows:** Guaranteed Income (Pensions, Wages, Annuities) + Social Security + Non-taxable Income + **Required Minimum Distributions (RMDs)**.
- **Cash Outflows:** Desired Living Spending + Additional Milestone Expenses + Base Taxes (tax on pensions, RMDs, and SS) + Other Scheduled Taxes.
- **Net Cash Flow = Total Inflows - Total Outflows.**

2. Handling Cash Surpluses:

If Net Cash Flow is positive, your guaranteed income and RMDs exceed your living expenses and taxes. You have a surplus! No discretionary portfolio withdrawals occur. The unspent surplus is automatically deposited into your **Taxable Brokerage Account**, where it begins earning returns.

3. The Withdrawal Priority Waterfall (Covering Deficits):

If Net Cash Flow is negative, you have a cash deficit. The application pulls funds from your investment accounts according to a strict, hard-coded withdrawal sequence:

1. **Tier 1: Taxable Brokerage & Cash:** Withdrawn first dollar-for-dollar. No income tax or penalties are assessed.
2. **Tier 2: Pre-Tax IRAs and 401(k)s:** Tapped only when taxable accounts are empty. Tapped proportionally between spouses. Processed through the gross-up solver to cover the extra taxes and penalties triggered by the distribution.
3. **Tier 3: Roth IRAs & Roth 401(k)s:** Tapped only after pre-tax accounts are completely exhausted. Withdrawn 100% tax-free and penalty-free.
4. **Tier 4: Health Savings Accounts (HSA):** User HSA first, then Spouse HSA. If marked for medical, withdrawn tax-free. If marked non-medical, grossed up for taxes and penalties.

4. Spousal Rollover Mechanics:

Upon the death of the first spouse, the surviving spouse automatically inherits the deceased spouse's remaining Pre-Tax and HSA balances via spousal rollover, combining them into the survivor's accounts without triggering immediate taxable liquidation.

CRITIQUE & WEAKNESS: WITHDRAWAL WATERFALL

Criticisms & Weaknesses of the Withdrawal Strategy:

- **Rigid Conventional Waterfall:** Spending Taxable first, then Pre-Tax, then Roth is the textbook default, but it is often mathematically sub-optimal. Allowing pre-tax accounts to grow untouched until age 73 or 75 creates an explosive 'RMD Tax Bomb', forcing huge mandatory withdrawals that push retirees into peak tax brackets.
- **Lack of Strategic Roth Conversions:** The app cannot simulate converting pre-tax assets to Roth at low tax brackets during the early retirement 'tax valley' (between retirement and Social Security/RMDs).
- **No Dynamic Spending Rules:** Real retirees cut discretionary spending during bear markets (Guyton-Klinger guardrails). The model assumes you spend the exact same rigid dollar amount regardless of portfolio losses.

Section (f): How Asset Growth Is Calculated and Used

Asset growth represents the engine of wealth accumulation and portfolio sustainability. How growth is compounded, when contributions and withdrawals occur, and how market correlation is modeled are critical to the accuracy of the simulation.

1. The Annual Growth Sequence (Order of Operations):

For each account in each year, the software executes an exact three-step calculation:

- **Step 1: Add Scheduled Contributions:** $\text{Balance}_{\text{before}} = \text{Beginning Balance} + \text{New Contributions} + \text{Any Life Insurance Proceeds}$.
- **Step 2: Apply Investment Growth:** $\text{Growth} = \text{Balance}_{\text{before}} \times \text{Annual Return Rate } (r_t)$.
- **Step 3: Mid-Year Balance:** $\text{Balance}_{\text{mid}} = \text{Balance}_{\text{before}} + \text{Growth}$.

Key Takeaway on Timing: New contributions are added at the start of the year and receive a full year of investment growth. Required distributions and living expense withdrawals are deducted at the end of the year, after investment growth has already been credited. If an account balance is negative, investment growth is set to 0.0 (no interest or debt compounding occurs).

2. Returns in the Deterministic Projection:

In the Deterministic Projection, the annual return rate (r_t) is constant and unchanging, equal to the expected mean return you entered for each account (e.g., exactly 6.0% every year).

3. Returns in the Monte Carlo Engine (Correlated Multi-Asset Modeling):

In the real world, if stocks crash, your IRA and your spouse's IRA don't move independently—they plunge together. The application models this using a sophisticated **Cholesky decomposition factor model** based on three correlated foundational asset classes:

- **Stock / Bond Correlation:** -0.10 (mild flight-to-safety diversification).
- **Stock / Cash Correlation:** 0.00 (uncorrelated).
- **Bond / Cash Correlation:** +0.20 (modest positive correlation).

For each account, the application automatically infers its underlying asset allocation from its mean return (e.g. $\geq 7\%$ = 100% Stock; 4% to 7% blends Stocks & Bonds; 2.5% to 4% blends Bonds & Cash). When market shocks occur, all accounts share the same underlying market factors weighted by their asset allocation. This ensures realistic cross-account correlation while strictly preserving each account's user-specified expected return (mean) and volatility (standard deviation).

CRITIQUE & WEAKNESS: ASSET GROWTH MODEL

Criticisms & Weaknesses of the Asset Growth Engine:

- **Static Asset Allocations (No Glidepath):** The model assumes you hold the exact same asset allocation forever (from age 60 to 95). Most financial plans reduce equity exposure as retirees age (a bond tent or declining equity glidepath) to mitigate sequence risk.
- **Normal Distribution Assumption (Fat-Tail Blindness):** The Monte Carlo engine draws from normal (bell curve) distributions. Actual financial market returns exhibit 'fat tails' and extreme skewness (crashes like 1987 or 2008 occur far more frequently in reality than standard normal statistics predict).
- **End-of-Year Withdrawal Advantage:** Liquidating withdrawals strictly at the end of the year allows portfolios to earn a full year of market growth on money that was actually spent throughout the year (e.g., January groceries), slightly overstating portfolio performance.

Summary Matrix: Key Architectural Weaknesses & Professional Recommendations

The following matrix summarizes the critical areas identified in this evaluation, their practical impact on your retirement plan, and recommended solutions:

Component	Identified Weakness / Omission	Practical Impact on Retiree	Recommended Enhancement
Success Metric	Success measured only at terminal year (age of death); interim deficits accrue at 0% interest and can be erased by later income.	Overstates plan safety. Retirees who go broke for several years in mid-retirement can be mistakenly categorized as 'successful'.	Implement continuous insolvency tracking: flag a run as failed if wealth drops below zero at <i>any</i> point in the simulation.
Tax Engine	0% capital gains tax on taxable account withdrawals; no cost-basis tracking; missing Medicare IRMAA tiers.	Understates retirement taxes by thousands of dollars annually, especially for retirees with substantial taxable brokerage assets.	Add capital gains tax estimation (e.g., 15% on 50% gain ratio) and Medicare IRMAA income-tier surcharges.
Withdrawal Strategy	Rigid conventional hierarchy (Taxable → Pre-Tax → Roth). No dynamic Roth conversion capability.	Creates an unnecessary 'RMD Tax Bomb' in the mid-70s. Leaves retirees paying significantly more lifetime taxes than necessary.	Add a tax-bracket-filling withdrawal strategy and proactive annual Roth conversions up to the 12% or 22% bracket ceiling.
Historical Stress Test	Post-2024 calendar wrap-around to 1926; deterministic identical shocks across all runs during crisis window.	Multi-year stress tests starting in recent years can create bizarre historical combinations (e.g., 2024 straight into 1926).	Cap historical sequences at 2024 and transition into stochastic returns rather than wrapping around to 1926.
Asset Growth & Returns	Normal distribution draws ignore market 'fat tails'; static asset allocation ignores age-based de-risking glidepaths.	Underestimates the probability of severe multi-standard-deviation market crashes; assumes retirees never adjust risk tolerance.	Incorporate Student's t-distribution or historical resampling (bootstrap) and allow age-based equity glidepaths.

Conclusion

The retirement simulation application represents a remarkably sophisticated, high-performance planning engine. Its integration of Numba JIT compilation allows it to simulate tens of thousands of complex tax, cash-flow, and market paths in a fraction of a second. Its implementation of SECURE 2.0 RMD ages, post-2025 federal tax brackets, spousal rollovers, correlated multi-asset shocks, and IRS Social Security formulas places it far ahead of typical consumer retirement calculators.

However, for robust real-world decision-making, users must remain mindful of the critical weaknesses highlighted in this report: primarily the terminal-only success definition (which forgives mid-retirement insolvency), the omission of capital gains taxes on taxable brokerage liquidations, and the rigid withdrawal sequence that leaves portfolios exposed to the RMD tax bomb. By understanding both the mathematical mechanics and the built-in assumptions, retirees and advisors can use these results with confidence, treating them not as an infallible crystal ball, but as a disciplined, stress-tested stress simulator for lifelong financial security.